MIL-I-6514B(USAF)
15 March 1988
SUPERSEDING
MIL-I-6514A(USAF)
8 June 1956

MILITARY SPECIFICATION

INDICATOR, OIL PRESSURE, 0-200 PSI, 320 DEGREE, TYPE B-9C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating, synchro, oil pressure indicator, designated Type B-9C.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

MIL-STD-129	Marking For Shipment And Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MIL-STD-810C	Environmental Test Methods
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements.

MS STANDARDS

MS33585	Pointer, Dial, Standard Design Of Aircraft Insturment

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadephia PA 19103.)

(Nongovernment standards and other pubications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in Specification MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design and construction.

3.2.1 Dual indicator. The indicator shall be a dual indictor.

3.2.2 Indicator dial. The indicator dial shall conform to figure 1.

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3.2.3 Visibility of dial markings. All graduations, numerals, pointer, and other markings on the dial shall be readable from any point within a frustum of the cone, the side of which makes an angle of not less than 30 degrees with a perpendicular to the dial, and the small diameter of which is the aperture of the dial.

3.2.4 Pointer. The shaded portion of the MS33585 pointer shall be finished in fluorescent-luminescent material in accordance with Specification MIL-L-25142.

3.2.4.1 Pointer movement. No stops shall be incorporated in the indicator mechanism to limit pointer movement. The pointer shall be free to rotate through 360 degress of an arc clockwise or counterclockwise from the zero graduation.

3.2.5 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the indicator. They shall be No. 6-32 NC-2 round head brass machine screws having a durable dull black finish and of sufficient length to permit mounting on a panel 0.188 inch thick.

3.2.5.1 Envelope. An envelope containing mounting screws shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.3 Weight. The weight of the indicator shall not exceed 1.25 pounds.

3.4 Markings.

3.4.1 Fluorescent-luminescent markings. The following markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as shown in table I.

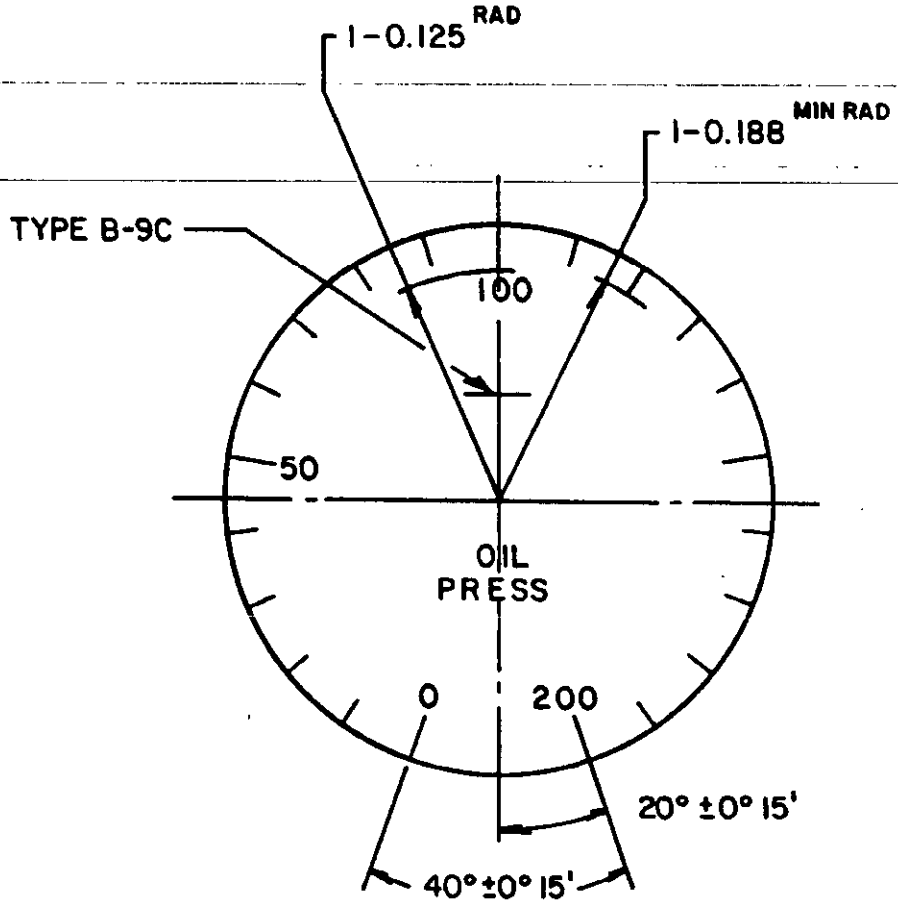
TABLE I. Dimensions of fluorescent-luminescent markings.

Marking	Height or length inch ±0.016	Width of line or graduation inch ±0.005
Numerals 0, 50, 100 and 200	0.250	0.031
50-pound graduations	0.250	0.031
10-pound graduations	0.188	0.016
Lettering "OIL PRESS"	0.125	0.020

3.4.2 Durable dull black. "TYPE B-9C" shall be permanently and legibly marked on the dial in 0.047 inch letters. This marking and all other markings not otherwise specified shall be finished in durable dull black.

3.5 Identification of product.

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DIMENSIONS IN INCHES.

FIGURE 1. Dial.

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3.5.1 Nameplate. The nameplate shall be marked with the following information:

INDICATOR, PRESSURE, OIL
Type B-9C
Specification MIL-I-6514A
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or order No.
Stock No.
Manufacturer's name or trade-mark
US

3.6 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirement.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicted or actual, nor does it commit the Government to acceptance of defective material.

4.1.1 General. The sampling, inspection, and test procedures shall be in accordance with Specification MIL-I-7057, and as specified herein:

4.2 Acceptance tests.

4.2.1 Individual tests.

4.2.1.1 Scale error and friction. The difference between the first and second readings shall not exceed the friction tolerance specified in table II. The second reading shall not exceed the scale error tolerance specified in table III.

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TABLE II. Oil pressure indicator.

Test	Tolerance
Friction:	
Room temperature	2.0
High temperature	2.0
Cold temperature	3.0
Humidity	3.0
Position error	0.5
Vibration:	
Pointer oscillation	1.5

TABLE III. Oil pressure indicator.

Psi	Tolerance
0	1.0
50	1.0
100	1.0
150	1.0
200	1.0

4.2.1.2 Position error. The Position error test shall not exceed the amount specified in table II.

4.2.2 Sampling tests.

4.2.2.1 Low temperature. Friction shall not exceed the tolerances specified in table II. The scale error shall not exceed the tolerances specified in table III.

4.2.2.2 High temperature. Friction shall not exceed the tolerances specified in table II. Scale error shall not exceed the tolerances specified in table III.

4.2.2.3 Vibration. The pointer oscillation and variation shall not exceed the tolerances specified in table II.

4.2.3 Special tests.

4.2.3.1 Rain. The Rain test shall be in accordance with MIL-STD-810C Method 506.1, Procedure II, except the indicator shall be mounted for 0.500 hour with the dial in a vertical plane and 0.500 hour with the dial in a horizontal plane (dial face upward) in the rain test chamber. At the completion of the test, the indicator shall be examined, and there shall be no evidence of water penetration or damage.

4.2.3.2 Low temperature exposure. Tests for Scale error and friction as specified in 4.2.1.1 shall be at the points specified in table III. Friction shall not exceed the tolerances specified in table II. Scale error shall not exceed the low temperature tolerance specified in table III.

4.2.3.3 High temperature exposure. Tests for Scale error and friction as specified in 4.2.1.1 shall be at the points specified in table III. Friction shall not exceed the tolerances specified in table II.

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4.2.3.4 Humidity. The Humidity test shall be as specified, except that the temperature shall be 65° ±2°C and for only 5 cycles. Upon completion of the test, the indicator shall meet the requirements specified for room temperature Scale error and friction tests as specified in 4.2.1.1.

4.2.3.5 Dust (Fine Sand). The indicator shall be subjected to the dust test per MIL-STD-810C, Method 510.1, Procedure I. At the end of this test, the indicator shall be subjected to the scale error and friction tests (4.2.1.1) at room temperature. Scale errors and friction shall not exceed the tolerances specified in tables II and III for room temperature.

4.2.3.6 Salt fog. The indicator shall meet the requirements specified for room temperature Scale error and friction tests as specified in 4.2.1.1. No damage to the indicator shall result from this test.

4.3 Inspection of packaging. Tests of methods of packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of the specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, on Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indictor shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure indicator does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indictor shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2)

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5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type B-9C indicator covered by this specification is intended to indicate the pressure in the lubricating system of aircraft engines.

6.2 General. The instructions contained in section 6 of Specification MIL-I-7057, including Provisions for Preproduction tests, are applicable to and form a part of this specification.

6.3 Subject term (key word) listing.

Indicator
MIL-I-7057
Oil
Pressure
Synchro
Type B-9C
C54
C131
200 PSI
320 Degree

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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